

The MySQL team at Sun Microsystems is planning to bring significant changes to Common Table Expressions, Windowing functions, invisible indexes, and roles for user privileges. Likewise, MySQL 8 will bring features like Group Replication and Document Store from major server releases.

Sun Microsystems has skipped MySQL v6 and v7 for the eighth iteration. The MySQL v6 release was in the pipeline before the acquisition of MySQL AB. But the new parent company did not develop it further for production use.

You need to wait for some time to experience MySQL 8. Meanwhile, you can keep yourself informed on all the changes through the official release notes.

Google hosts 'Operation Rosehub' to patch thousands of open source projects

To narrow down the impact of the infamous 'Apache Commons Collections Deserialisation Vulnerability' that had affected several Java-based programs, Google has silently kickstarted its 'Operation Rosehub'. A 50-member team of Google



employees drove the new initiative for the first time in 2016 to help patch over 2,600 open source projects.

Internally called 'Mad Gadget' by Google's engineers, the vulnerability was spotted in early 2015. It grabbed the attention of companies like Oracle, Cisco,

Red Hat, Jenkins, VMware, IBM, Intel, Adobe and HP a few months after its discovery by security researchers from Foxglove Security. The IT companies immediately issued security alerts to let enterprises patch their proprietary offerings. However, a Google employee took the step to fix the issue in affected open source projects.

"Operation Rosehub was organised from the bottom-up on companywide mailing lists. Employees volunteered and patches were sent out in a matter of weeks," Google's software engineer Justine Tunney wrote in a blog post.

The researchers had reported that the vulnerability was a part of the seven 'gadget' classes within the Apache Commons Collections library versions 3.0, 3.1, 3.2, 3.2.1 and 4.0. These were the classes for handling Java object deserialisation that was used alongside the serialisation function to convert data from one format to another.

As the library was vital for many software operations, it was deployed by various commercial and open source projects. The Google employee used the pull request feature on GitHub to inform developers to patch their community solutions at the initial stage. But to reach the masses, a new development was needed.

Google has the Rosie tool that helps developers implement large-scale changes to codebases owned by its engineering teams. But GitHub, where a large number of Mad Gadget-affected projects were hosted, does not offer any such help. This is the reason the search giant formed a special task force and started working on the patches. "Patches were sent to many projects, avoiding threats to public security for years to come," said the Google engineer.

Though some of the patches were just one-line changes, Google's engineers took months to fix the vulnerable projects on GitHub. They spent a part of their daily routine at Google to jointly work on the patches.

For the remaining projects that are yet to be patched, the Google team is using an open source data set on BigQuery. This helps the engineers to identify the vulnerability in listed solutions.

"Going forward, we believe the best thing to do is to build awareness. We want to draw attention to the fact that the tools now exist for fixing software on a massive scale, and that these work best when that software is open," Tunney concluded.

Pixel 2.0 brings Arduino to wearables world

Kickstarter project Pixel 2.0 has expanded the usability of the Arduino platform by bringing it to the wearables market.

The square-shaped Pixel 2.0 is an Arduino Zero compatible board with an OLED panel on one side. The 128×128 colour display is 3.8cm (1.5 inch) in size, which makes it an easy fit for wearables. You can use the board either like a smartwatch or embed it into a jacket to make it a smart wearable.

For performing different tasks on the small screen, the Pixel 2.0 has a 32-bit 48MHz Atmel ATSAM21G18 Cortex M0+ microcontroller along with 32K of RAM. There is also a microSD card slot that gives you the source to load your custom-made graphics and text.

The 4.6cm (1.8 inch) board has 14 digital I/O pins, six analogue input pins and an output pin. It comes preloaded with Arduino/Genuino Zero bootloader, and is compatible with the Arduino's SPI and SD libraries. You can also use Adafruit's graphics library to easily create new images and drawings for the Pixel 2.0.

Though the pre-installed bootloader is specific to Arduino developments, there is a standard SWD header on the Pixel that has full compatibility with Atmel ICE.



Unlike some other Arduino compatible boards, the Pixel 2.0 is entirely an open source model. This means that you will get access to its schematics and PCB layout. Moreover, you can modify the Pixel design and even integrate it into your own boards — without the need for any prior royalties and licensing fees.

The Pixel 2.0 is available at US\$ 75 on Kickstarter, with shipment scheduled to begin in June this year.